

⚡ Feature Engine Pro

Automated Feature Selection — Audit Report

📄 Executive Summary

Generated on 2026-05-01 14:22:21 • Pipeline runtime: 35.77s

34

Input Features

4

Selected Features

89

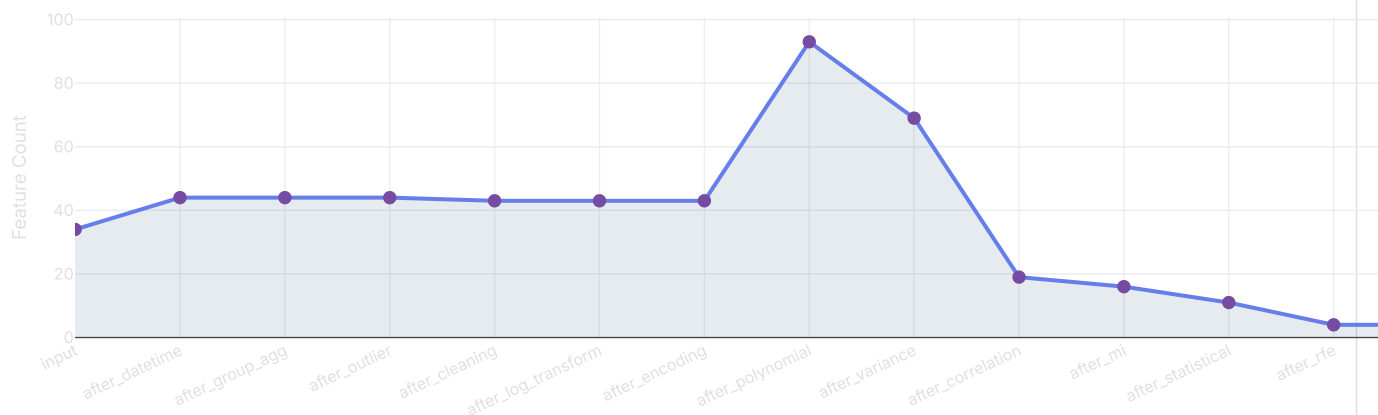
Features Dropped

95.7%

Dimensionality Reduction

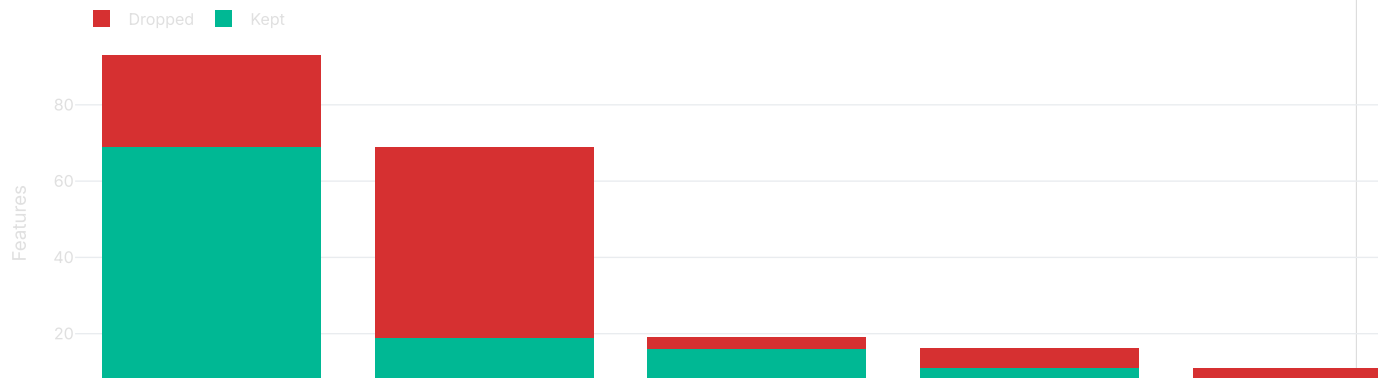
Column Journey

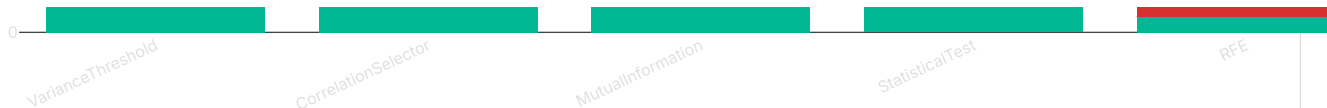
Feature count at each pipeline stage — shows where features were created and destroyed



📊 Feature Filtering Funnel

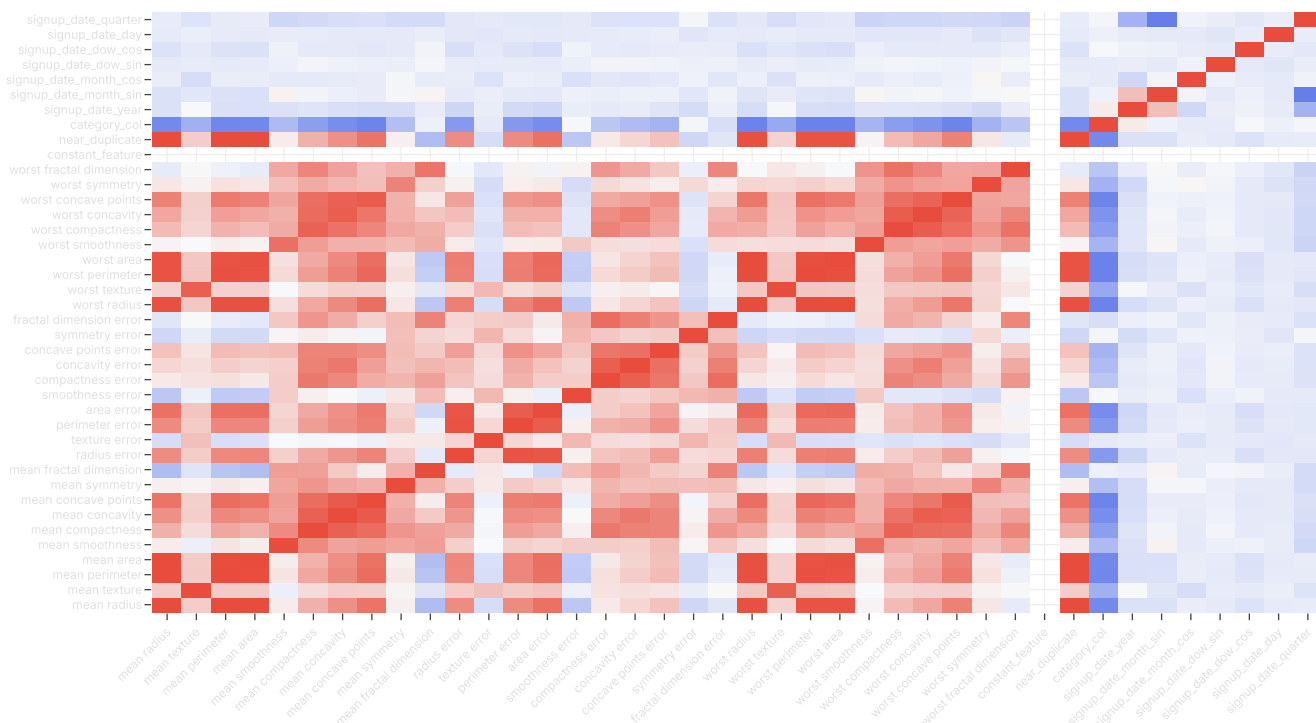
How features were kept or dropped at each pipeline stage





Correlation Matrix

Interactive heatmap — hover for details, scroll to zoom



Performance Impact

Proving that fewer features = same or better performance

93

Features Before

4

Features After

95.7%

Reduction

Metric	Before Selection	After Selection	Change
accuracy	0.9978 ± 0.0044	0.9978 ± 0.0044	+ 0.0000
f1_weighted	0.9978 ± 0.0044	0.9978 ± 0.0044	+ 0.0000

Complete Audit Trail

Every feature decision with mathematical justification — search and filter below

🔍 Search features, stages, or reasons...

All (93)

✓ Kept (4)

✗ Dropped (89)

Feature ↑	Status ↑	Stage ↑	Mathematical Reason ↑
mean radius	✓ KEPT	VarianceThreshold	Variance 11.065510 above threshold 0.01.
mean texture	✓ KEPT	VarianceThreshold	Variance 16.572300 above threshold 0.01.
mean perimeter	✓ KEPT	VarianceThreshold	Variance 523.432372 above threshold 0.01.
mean area	✓ KEPT	VarianceThreshold	Variance 90536.810847 above threshold 0.01.
mean smoothness	✗ DROPPED	VarianceThreshold	Variance 0.000180 below threshold 0.01.
mean compactness	✗ DROPPED	VarianceThreshold	Variance 0.002503 below threshold 0.01.
mean concavity	✗ DROPPED	VarianceThreshold	Variance 0.004361 below threshold 0.01.
mean concave points	✗ DROPPED	VarianceThreshold	Variance 0.001355 below threshold 0.01.
mean symmetry	✗ DROPPED	VarianceThreshold	Variance 0.000687 below threshold 0.01.
mean fractal dimension	✗ DROPPED	VarianceThreshold	Variance 0.000042 below threshold 0.01.